

CONTINUATION — HelixTranslate

session-resumption file

Revision: 16 **Last modified:** 2026-06-13T00:00:00Z **Purpose:** Single canonical out-of-the-box entry point for any fresh session (§11.4.131 / §12.10 / §11.4.127). To resume: point a new session at THIS file, run `git fetch --all`, and say **continue**.

Session wrap-up — W13, W15 (api/grpc), W16 all resolved (HEAD 1cc66ac)

- **W15 api/grpc DONE (real, evidence-backed)** — subagent-driven. Built build-tagged integration suites: `pkg/api/server_realhttp_integration_test.go` (real HTTP via `httptest` + real JWT + real Postgres via `brokertest`: `health`, `JWT 401/200`, `login→token` round-trip, / `api/v1/verified-models`, `session persist` round-trip) and `pkg/grpc/server_storage_integration_test.go` (real gRPC over `bufconn` + real Postgres: `submit→persist`, `status`, `cancel`, `list-count`, `8× concurrent`, `error-code` via `status.FromError`; §1.1 mutation-proven). LLM-translate leg = honest §11.4.3 SKIP (no provider key). Normal `go test` boots NOTHING; SKIP-clean without `podman`.
- **REAL AUTH BUG fixed (W15 api leg, commit 0e2a999)** — `InMemoryUserRepository.Create` stored the caller's pointer; `CreateUser` clears `Password=""` post-Create → blanked the stored bcrypt hash → EVERY user registered via `CreateUser` could never log in. Fix: repo stores a COPY. Conductor-reproduced RED(401)→GREEN(200) on `TestAPIRealHTTP_LoginTokenRoundTrip` (§11.4.115). Remaining W15: only the pure LLM-translation E2E (operator-gated on a real provider key).
- **W13 DONE (commit 1cc66ac)** — 8 PascalCase `.gitmodules` section labels → lowercase (paths were already compliant). Done §9.2 backup-first; all 11 submodules verified resolving to identical baseline SHAs after `git submodule sync`, dirty challenges/helix_qa preserved, build exit 0, backup removed post-gate.
- **W16 RESOLVED = KEEP Models/ as-is** — the `pkg/modelsbridge` bridge already consumes `digital.vasic.models` (functional need met). Converting `Models/` → a git submodule requires creating a NEW upstream repo under an owned org — an operator-gated repo-creation action (§11.4.101 block-only-when-irreversible+undeterminable; §11.4.122 keeping is the safe no-change path). Convert path remains available on operator request.

Autonomous-safe queue now EMPTY. Genuinely-remaining = operator-gated only: W15 LLM-translation E2E (real provider key) + W16 convert (new upstream repo decision). Per §11.4.94 this is legitimate idle.

LLMsVerifier real-key integration — DONE (HEAD e7d6d67; submodule 95fb1cce)

Operator directive: use `~/api_keys.sh` so the LLMsVerifier submodule performs its mandatory steps and provides valid providers + models to the System. **Done, evidence-backed (§11.4.83 docs/qa/llmsverifier_20260613_182600/, key-free):** - Ran the submodule's REAL discovery/verification with real keys → **89 verified models** (DeepSeek 2, Cerebras 2, Groq 16, Mistral 69; OpenAI/Anthropic honestly fail — no key in `api_keys.sh`). - **Two real bugs found + fixed, both §11.4.135 mutation-proven guards (exact production errors):** - **Bug B (main, internal/verifier/client.go) load-bearing:** the System's SSOT client couldn't decode the server's `{"count":N,"models":[...]}` envelope → consumed ZERO models while server healthy. Fixed (envelope-aware decode). Guard: `client_envelope_test.go`. System now fetches all 89 e2e. - **Bug A (submodule llmverifier/models.go, 95fb1cce, pushed GitHub+GitLab):** numeric `context_window` (Groq) crashed the whole provider decode → 0 models. Fixed (tolerant `UnmarshalJSON`). Guard: `context_window_unmarshal_test.go`. - **Secrets:** keys sourced inline only, never printed/logged/committed (§11.4.10); evidence dir + edits secret-scanned clean; throwaway run artifacts removed (§11.4.14).

Honest follow-up gaps (tracked, not bluffed — §11.4.6): - (G1) Submodule `verify` (report) pipeline does NOT persist into the server SQLite DB — the served DB was seeded via live discovery (verifier's own client+database pkgs). Architectural bridge worth a workable item. - (G2) Full ~30-provider exhaustive per-model coding-capability scoring timed out at 600s in one pass; reliability-scored the seeded set. A bounded/batched sweep is the follow-up. - (G3) Providers without keys (OpenAI/Anthropic/Perplexity/Together) fail auth honestly; add their keys to `api_keys.sh` to include them.

SHORT resumption sentence

Read `docs/CONTINUATION.md` then `.remember/remember.md`, run `git fetch --all`, and **continue** the HelixTranslate “raise-to-enterprise” mandate at **Phase 4 (tests/Challenges/HelixQA to ~100% per type + expand real ebook-translation evidence)** while finishing the Phase-3 tail (Models decision, governance gaps). Binding: anti-bluff §11.4 (real captured evidence, no false PASS), no-force-push §11.4.113, flat `snake_case` submodules §11.4.28/§11.4.29, every change independently reviewed §11.4.142, containers-first infra §11.4.76, release tags prefixed `helix_translate-` (§11.4.151). Work subagent-driven (§11.4.70) with ≥ 3 parallel streams (§11.4.103).

Live state anchors (moment-valid)

- **Parent HEAD:** see `git log` (W14b adoption commit via `commit_all.sh`) on main; pushed to BOTH `milos85vasic/Translator` + `HelixDevelopment/HelixTranslate` (verified, fast-forward, no force). Session commit chain: `b3dc7f9(llm_provider)` → `e775088` → `8a6af67` → `7331c54` → `76677aa` → `18c5137` → `de256dd` → `9004477` → `13d5e30` → `af1ef47` → `a9b38fa` → `218bfa0` → `1f06bf5` → `a0930fc` → `3027da2` → `b2377af` → `6d4ae47`.

- **llm_provider HEAD:** b3dc7f9 (W7 CONST-036 propagation block + regenerated html/pdf), pushed to HelixDevelopment/LLMProvider master.
- **Constitution HEAD:** 5e671fe (§11.4.151 added), pushed to all 6 upstreams; parent pointer bumped.
- **Build:** go build ./... = EXIT 0. **Total test coverage = 50.7%** (go tool cover -func total; see docs/testing/coverage_matrix.md).
- **Submodules (flat, snake_case):** challenges containers constitution doc_processor docs_chain helix_qa llm_orchestrator llm_provider llms_verifier security vision_engine.
- **Release prefix:** HELIX_RELEASE_PREFIX=helix_translate (.env git-ignored, .env.example tracked). First RC: helix_translate-1.0.0-dev-0.0.1 — create ONLY when genuinely green (§11.4.40), operator-gated.
- **Safety backup:** ../helix_translate_gitbackup_20260611T181545Z/repo.git.mirror (pre-rename 311c585).

DONE

Phase 1 — flat snake_case submodules (reviewed GO, pushed): 8 submodules renamed; Go pkg challenges/→pkg/challenge_runner/; docs_chain added; .env release-prefix. go build ./... EXIT 0. **Phase 2 — constitution §11.4.151** release-prefix rule → pushed to all 6 constitution upstreams; parent pointer bumped; inheritance meta-test PASS. **Phase 3 (partial):** - Security/ plain dir → flat security submodule (proven byte-identical to upstream, zero loss; §11.4.124). - All submodules + main synced to every upstream (one stale containers gitlab mirror FF'd). - **docs_chain OPERATIONAL:** .docs_chain/contexts/ tracked_docs.yaml registered; real pandoc+weasyprint sync produced docs/ CONTINUATION.{html,pdf} + README.{html,pdf}; verify = in-sync. See docs/ DOCS_CHAIN.md. - **Real translation PROVEN (anti-bluff):** EN→Serbian via DeepSeek deepseek-chat, 208 Cyrillic chars, differs from source, no placeholders. Evidence: qa-results/translation/<ts>/ (raw, git-ignored). Asset: test/assets/crow_and_pitcher_en.txt. - **Fixed a regression I introduced:** challenge scripts' PROJECT_ROOT="\${SCRIPT_DIR}/../.." broke when moved 1 level deeper → corrected to ../../..; 3 challenges re-verified PASS (cache_invalidation, model_verification_gate, anti_bluff step1-3).

CLOSED this session (2026-06-13, batch e775088 / llm_provider b3dc7f9 — all pushed, captured evidence)

- **W4**  pkg/security flaky Short-TTL — widened TTL→1min; 50× + -race + CPU-load all PASS.
- **W5**  pkg/distributed 120s timeout — bounded SMTP send (real defect) + in-process test servers; 3× ~48.5s, vet 0.
- **W6**  internal/verifier/discovery 600s timeout — injectable URLs (prod unchanged) + sentinel stub + honest SKIP; 3× ~1s, paired §1.1 mutation proven.
- **W7**  llm_provider CONST-036 propagation block — challenge GREEN; html/pdf regenerated.
- **W8**  anti_bluff_execution_challenge.sh BSD-sed — portable tmpfile sed (§11.4.67); isolated rc=0 proof.

- **W9** ✓ tracked .bak/.backup residue — git-history-investigated (§11.4.124), removed + .gitignore; build 0.
- **W10** ✓ -script latin→Cyrillic — pkg/script was never imported into unified-translator; normalizeScript wired; RED→GREEN 7 tests + 3×.
- **W12** ✓ 8 challenge scripts brittle root — git -C rev-parse --show-toplevel w/ fallback (§11.4.111); all parse:OK root:OK.
- **D2** ✓ (wave2, 7331c54) test/unit intermittent FAIL — TestVerifier/EventEmission timing race (50ms Sleep vs goroutine-dispatched handler) + data race (proven -race). Mutex eventCollector + bounded poll. BEFORE 1/10 FAIL+race → AFTER -race -count=10 10/10 PASS.
- **W2-A** ✓ (wave2) cmd/unified-translator coverage 3.2%→18.0%, mutation-verified; + latent dangling-pointer fix (Steps []TranslationStep→[]*TranslationStep).
- **W2-B** ✓ (wave2) pkg/storage coverage 32.9%→35.3% (Redis cache-key helpers 0%→100%, real SQLite round-trips), mutation-verified, daemon paths honest-SKIP.
- **wave3** (18c5137): **W3** pkg/translator/llm stress/chaos + **FIXED real prod data race** in BaseTranslator cache+stats (RWMutex, -race ×3); **W2** sshworker 19.6%→39.7%; **W2** batch 77.2%→90.6%.
- **wave4** (de256dd): **sshworker progress-map leak** on ctx.Done() fixed (defer cleanup, RED 300/300→GREEN -race); **D3** pkg/translator{,/llm} test-only EventBus races fixed (asyncFlag); **D7** pkg/events TestEventBus_Subscribe race fixed; **W2** verification 64.0%→73.5%; **W2** script (already 100% stmt) +11 behavioral/digraph tests.
- **D5** (9004477) **FIXED real bug** pkg/verification parseNote: CONTENT-only note (no IMPLICATIONS) silently dropped → finalize Content; RED→GREEN, mutation-verified.
- **D6** (9004477) **FIXED** verified_factory map-order flake → order-independent ElementsMatch; 10/10.
- **wave5** (af1ef47): **W14** scripts/commit_all.sh wrapper (no-force/multi-upstream/FF-only/quiescence/explicit-paths/bg-push) + hermetic selftest 13/0 (conductor-run verified) — NOT yet wired to real remotes (line-by-line review pending); **W2** format 0%→97.6%, markdown 73.9%→80.1%, coordination 0%→92.9% + **GENUINE nil-guard fix** in TranslateWithRetry (mutation-verified: revert→nil-panic). §11.4.147: format+coordination were rate-limit-crashed at report stage; conductor verified+adopted their work (not lost, not blindly trusted).

Session total: ~39 items resolved (+ D10 pkg/distributed FULLY -race-clean 9→0 races [4 sources incl. FallbackManager.Stop() lifecycle fix]; + D9 hardware parser testability 44.6→68.4%)

Prior: ~37 items (+ models bridge wiring digital.vasic.models into the system; + D12 REAL CORS auth-bypass vuln fixed; + D11 doc)

Prior: ~34 items (+ W15: real Postgres+Redis on-demand integration proven; PG connection-pool DoS bug found+fixed; containers StartRedis helper added)

Prior: 31 items (+ wave7: grpc 0→53.3% bufconn, api 55→62.2% httpstest, security +14 adversarial tests — all real-protocol/anti-bluff, auth confirmed hardened)

Prior: 29 items (+ wave6: websocket race-fix, hardware/distributed chaos tests, D10#2 api_logger PRODUCT data-race fix)

Earlier session total: 26 items (W4-W10,W12 + D2 + W2×6[unified-translator/storage/batch/sshworker/verification/script] + W3 translator + sshworker-leak + D3 + D7 + D5 + D6). Real prod bugs fixed: translator cache race, SMTP no-deadline, dangling step-ptr, sshworker map-leak, event races (D2/D3/D7), note-content loss (D5).

OPEN workable items (queue — many parallelizable, disjoint scope)

#	Item	Type	Notes / evidence
W1	Models/ (digital.vasic.models) orphan module	RESOLVED- via-bridge	Operator directed “create bridge”: pkg/modelsbridge now CONSUMES digital.vasic.models (go.mod require+replace=>./Models), contract-tested 100%, mutation-verified (146b2b5). No longer orphan. W16 (convert to git submodule) still optional/operator-gated. Done: unified-translator 18%, storage 35.3%, batch 90.6%, sshworker 39.7%, verification 73.5%, script(100%+depth), translator/llm(+stress/chaos). Also grpc 53.3% (bufconn), api 62.2% (httptest), security 84.8% +adversarial, hardware/distributed chaos. NOTE: gRPC/api-server/server/redis/postgres at 0% are integration-infra-gated (need real daemons) — W15 containers-first unlocks these; honest deferral per package.
W2	Coverage → ~100% (§11.4.27); ongoing	Task	
W3	Missing test types: chaos, ddos, scaling, ui, ux, full-automation (in-module)	Task	§11.4.85 chaos/stress mandatory.
W11	Wire more docs_chain contexts (Issues/Fixed/Status once they exist)	Task	docs_chain proven operational.
W13	.gitmodules section labels still PascalCase	Task (cosmetic)	Needs per-submodule .git/modules/* gitdir move.
W14		Task	

#	Item	Type	Notes / evidence
	No project commit wrapper scripts/commit_all.sh (§11.4.22)		IN-FLIGHT (wave5): multi-upstream push + no-force §11.4.113 + explicit-paths spec + quiescence + background-push §11.4.88 + hermetic self-test. Review before wiring to real remotes.
W15	Containers-first infra (§11.4.76)	Feature	storage DONE: real Postgres+Redis on-demand integration via containers/pkg/brokertest (StartPostgres + new StartRedis), build-tagged; pkg/storage CRUD+chaos exercised real DBs; found+fixed PG unbounded-pool DoS (default 25/5). REMAINING: api/grpc DB+LLM <i>pipeline</i> paths need operator-gated LLM keys/SSH (can't autonomously). Pattern + helpers in place for any future slice.
W16	Convert Models/ →models submodule after W1 resolved	Task	Blocked on W1.
D1	docs_chain + security CONSTITUTION.md lack literal CONST-036	Task	§11.4.118 discovery. They use §11.4.X cascade scheme (not CONST-NNN); challenge doesn't check them. Governance-design decision needed — do NOT blindly inject CONST-NNN (§11.4.6).
D4	pkg/script Latin→Cyrillic morpheme-boundary mis-transliteration	Bug (open)	конјугација→коњугација (should be коњугација); инјекција, надживети similar. Greedy digraph matching; real fix needs morpheme/dictionary model (high blast radius) — operator-aware decision. Pinned with KNOWN-LOSSY tests.
D8	✅ FIXED (218bfa0) pkg/markdown bare-leading --- data-loss	Bug (fixed)	Frontmatter now must begin at first non-blank line; later --- = HR/ chapter separator.

#	Item	Type	Notes / evidence
			RED(1ch)→GREEN(2ch), mutation-verified. Minor →dash numbering remains pinned/unfixed (D-list, low priority).
D9	✅ FIXED (fc66c62) hardware parser testability	Task	Extracted 12 pure parsers from exec callers (behavior-preserving); 44.6→68.4%, mutation-verified.
D10	3 baseline data races (FACT, captured -race)	Bug	#2 api_logger PRODUCT race ✅ FIXED (3027da2, snapshot, mutation-verified). #1 ssh_pool + #3 performance ConnectionPool. ✅ ALL FIXED (fc66c62): ssh_pool options API, perf locked-map + count-snapshot, FallbackManager.Stop() lifecycle. pkg/distributed now 0 races.
D11	✅ FIXED (b8a24f5) CLAUDE.md mislocated CORS under pkg/security	Task	Corrected: CORS is server-layer (cmd/server corsMiddleware + internal/config).
D12	✅ FIXED (b8a24f5) CORS auth-bypass vuln	Bug(security)	cmd/server corsMiddleware reflected ARBITRARY origin + Allow-Credentials:true under default ["*"] → any site could make credentialed cross-origin requests. Fixed: wildcard→literal * no-creds; creds only for specific allowlist. RED→GREEN, mutation-verified.
W14b	✅ DONE — reviewed + adopted	Task	Independent review: guards sound (force-reject full-argv, FF-only via merge-base -is-ancestor, multi-upstream URL-dedup, explicit-paths spec, quiescence). FOUND+FIXED cross-platform hardening: mutation grep lacked -I → GNU grep (Linux)

#	Item	Type	Notes / evidence
			false-aborts on binary .pdf/.html pathspecs (BSD grep doesn't; §11.4.81). Self-test 13/0. ADOPTED for normal source/doc commits. Dogfood surfaced finding #2: the §11.4.84 residue scan false-positives on files that legitimately CONTAIN marker strings (the script's own MUTATION_MARKERS def, governance docs, marker-referencing tests) — inherent to grep-based detection; §11.4.84 has no escape hatch (correct), so commit such files via manual explicit-path git. commit_all.sh is for the common case.

NEXT phases (priority order)

- **Phase 4** — drive W2–W8 + W10: per-type test suites to ~100% (§11.4.27), real Challenges + HelixQA bank execution with captured evidence, expand real ebook translations (all formats, whole chapters, anti-bluff §11.4/§11.4.69). Subagent-driven, ≥3 parallel streams.
- **Phase 5** — containers-first infra (W15); docs_chain context expansion (W11); full docs (guides/diagrams/SQL/templates) + enterprise Website.
- **Phase 6** — RC tag helix_translate-1.0.0-dev-0.0.1 across all submodules + main ONLY when genuinely green (§11.4.40), operator-confirmed.

Binding constraints

Anti-bluff §11.4 (real captured evidence, no false PASS) · no-force-push §11.4.113 (merge-onto-latest-main) · multi-upstream push §2.1 · flat snake_case §11.4.28/§11.4.29 · every change independently reviewed §11.4.142 · host-safety §12 (no suspend/poweroff) · no silent component removal §11.4.122 · containers-first §11.4.76 · release prefix §11.4.151 · subagent-driven default §11.4.70 · ≥3 parallel streams §11.4.103.

Gotchas

- /Volumes/T7 is **case-sensitive HFS+** (NOT the internal disk — verify on the actual volume, §11.4.6).

- No `scripts/commit_all.sh`; stage explicit paths, never `git add -A`; never `git add helix_qa` (nested third-party pointer drift).
- `PIPESTATUS` empty in this shell (zsh); use `$?` directly.
- Mutation challenges can leave `.go.backup/.bak` residue — scan + clean before commit (§11.4.84).